

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer-2023 Examination

Semester: 4
Subject Code: 203191251
Subject Name: Probability, Statistics and Numerical Methods

Date: 29/03/2023
Time: 2:00 pm to 4:30 pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions (Each of one mark)**(15)**

1. Write Trapezoidal rule for integration.
2. If H_0 is true and sample is rejected it is called
 (a) Random Variable (b) Type-1 error (c) Type-II error (d) None of these
3. In Newton's forward interpolation formula value of $p =$ _____
 (a) $\frac{x-x_0}{h}$ (b) $\frac{x_0-x}{h}$ (c) $\frac{x-x_n}{h}$ (d) None of these
4. If $r = 0.76$, $\sigma_x = 5.3$, $\sigma_y = 9$ then $b_{xy} =$ _____.
5. In case of fitting a quadratic curve $y = a + bx + cx^2$, which of the following is not a normal equation?
 (a) $\sum y = na + b \sum x + c \sum x^2$ (b) $\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$
 (c) $\sum xy = n \sum a + b \sum x^2 + c \sum x^3$ (d) $\sum xy = a \sum x + b \sum x^2 + c \sum x^3$
6. Which probability distribution is suitable for number of accidents take place at a particular spot?
7. "What is the probability that the sun will rise at 7 p.m.?" –Choose the correct answer from the following alternatives.
 (a) 0 (b) 0.5 (c) 1 (d) can not be predicted
8. Critical value of z at 5% level of significance in two - tail test is _____.
9. The correlation coefficient is independent of change of origin and scale (True/False).
10. Write the formula of Euler's method.
11. $\text{variance}(ax + b) =$ _____.
12. If two events A and B are independent events, $P(A \cap B) =$ _____.
13. If the mean of binomial distribution is 8 and $p = 0.5$ respectively, then $n =$ _____.
14. Write the relation between Regression coefficient and Correlation coefficient.
15. The degree of freedom in chi-square test when data is written in r rows and c columns _____.

Q.2 Answer the following questions. (Attempt any three)**(15)****A) Calculate the correlation coefficient between the following data**

x	5	9	13	17	21
y	12	20	25	33	35

B) Solve the initial value problem $\frac{dy}{dx} = x\sqrt{y}$, $y(1) = 1$ and hence find $y(1.5)$ by taking $h = 0.1$ using Euler's Method.

C) Solve $6x + y + z = 105$

$$4x + 8y + 3z = 155$$

$$5x + 4y - 10z = 65 \text{ by using Gauss-Seidel method.}$$

D) Use Newton's interpolation formula to find $f(1.3)$ from the following data

x	1	2	3	4
$f(x)$	1.1	4.2	9.3	16.4

Q.3 A) In a bolt factory machine, A, B, C manufacture 25%, 35% and 40% of the total output and (07)

out of the total manufacturing, 5%, 4% and 2% are defective bolts. A bolt is drawn at a random from the product and is found to be defective. Find the probability that it is manufactured from (i) Machine A (ii) Machine B, and (iii) Machine C.

B)

(1) If A and B are two events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cap B) = \frac{1}{12}$. Find (04)

(i) $P(A \cup B)$ (ii) $P(A/B)$ (iii) $P(B/A)$.

(2) The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively find $P(x \geq 1)$. (04)

OR

B)

(1) Find the second divided difference for the argument $x = 1, 2, 5$ and 7 for the function $f(x) = x^2$. (04)

(2) If a mean of Poisson distribution is 1.8 find (i) $P(x > 1)$ (ii) $P(x = 5)$. (04)

Q.4 A)

(1) A stenographer claims that he can write at an average speed of 120 words per minutes. In 100 trials he obtained an average speed of 116 words per minute with S.D. of 15 words. Is the claim justified? (Use 5% level of significance) (03)

(2) A die is thrown for 300 times and the following distribution is obtained. (04)

Number on the die	1	2	3	4	5	6
frequency	41	44	49	53	57	56

Can the die be regarded unbiased? (At 5% level of significance and 5 degree of freedom the value of $\chi^2_{5,0.05} = 11.07$)

OR

A) Derive the iterative formula of $\sqrt[3]{N}$ by Newton Raphson method and hence evaluate $\sqrt[3]{11}$. (07)

B) Evaluate $\int_0^3 \frac{1}{1+x} dx$ with $n = 6$ by using Simpson's 3/8 rule and hence calculate $\ln 2$. (08)